

Enterprise architecture for AI transformation

Why AI stalls at the enterprise, and the architecture that scales it past the pilot

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An architect's view of moving AI from scattered pilots to a governed capability.

Where this is heading

The pilot works. The enterprise is what breaks.

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The problem is not the model, it is the architecture around it

- The appetite and the budget are there: AI and big-data roles are the **fastest-growing to 2030** (WEF, 2025). Building a pilot has never been cheaper.
- Crossing from pilot to an enterprise capability is where it breaks:
 - Only about **48%** of AI projects reach production, and a prototype takes about **8 months** to get there (Gartner, 2024)
 - At least **30%** of generative-AI projects are expected to be **abandoned after proof of concept** by the end of 2025 (Gartner, 2024)
 - **More than 80%** of AI projects fail, around twice the rate of non-AI IT projects (RAND, 2024)
 - Gartner (2026) forecasts that **40%** of enterprises will **demote or decommission their AI agents by 2027** over governance gaps (a forecast, treat it as one)

A pilot proves the model can work. It says nothing about whether the enterprise can run it.

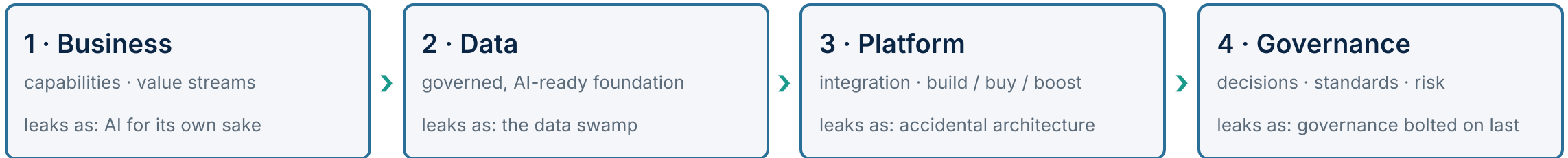
What actually blocks the crossing

The pilot ran on a borrowed dataset, one laptop, and nobody's standards. The enterprise version needs governed data, an integration path, a security and risk posture, an owner with a P&L, and a decision about what to build versus buy. None of that is a modelling problem.

Enterprise architecture is the discipline that turns a folder of pilots into one governed, fundable, scalable capability. The four layers that follow are where that crossing is won or lost.

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The architecture, end to end



Each layer has one job, one common way to lose the value, and one discipline that protects it. The next four slides take them in turn.

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1 • Business architecture: decide where AI earns its place

A map of business capabilities and value streams, so every AI initiative is tied to an outcome and an owner, not a demo.

Practical steps

- Start from the **capability map**: which capability does this initiative improve, and who owns its P&L?
- Frame each use case as a problem the business would pay to solve, with a measurable outcome
- Rank the portfolio by value and feasibility; fund a few, retire the rest
- Name an accountable owner before a line of code is written

The pitfall: AI for its own sake. Pilots launched because the technology is exciting, with no owner, no outcome, and no path to a budget. They demo well and die quietly.

What good looks like. Every funded initiative names the capability it serves, the owner who carries it, and the number that moves if it works.

Illustrative frames: business-capability modelling · value-stream mapping · a value-vs-feasibility portfolio view.

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2 · Data and information architecture: the governed, AI-ready foundation

A trustworthy, governed, modelled data layer, so every model downstream reads data with a known shape, owner, and meaning.

Practical steps

- Resolve core entities into **master records** with one authoritative key
- Write a **data contract** per dataset: schema, owner, freshness SLA, change policy
- Define the **semantic layer** once: entities, relationships, and metric formulae
- Attach governance, lineage, and classification at source, not at the report

The pitfall: the data swamp. Ungoverned data; every model silently inherits duplicate entities, ambiguous fields, and three versions of "revenue". Invisible at the pilot, expensive at scale.

What good looks like. A dataset is not "available" until it has a contract, an owner, and a place in the semantic layer. This is the layer that makes AI repeatable.

Illustrative patterns: Master Data Management · data contracts · the semantic layer · data-mesh domain ownership.

3 · Platform architecture: build, buy, or boost

The reference platform AI runs on, how it integrates with the estate, and the costed decision of what to build, buy, or augment.

Practical steps

- Set one **reference architecture** for how models are trained, served, integrated, and secured
- Make **build / buy / boost** a structured decision: weighted drivers and a risk-adjusted cost model, not a vendor pitch
- Define **fitness functions**: measurable tests that the architecture still performs as intended
- Plan integration to the systems of record (ERP, CRM, the data platform) up front

The pitfall: accidental architecture. Point solutions chosen one demo at a time: lock-in, duplicated capability, and an integration bill that arrives later.

What good looks like. One reference platform, a documented build/buy/boost decision per capability, and fitness functions that fail the build when a rule is broken.

Illustrative frames: reference architectures · MIT build-buy-partner · build/buy/boost decision models · fitness functions.

4 · Governance and the operating model: decisions, standards, and risk

The decision log, the standards, the security and risk posture, and the operating model that lets the rest run at enterprise scale without a committee in the loop.

Practical steps

- Record each significant choice as an **Architecture Decision Record**: context, options, decision, consequences, and a falsifier that would prove it wrong
- Run a **federated** operating model: central standards and guardrails, delegated delivery
- Build in security, data protection, and AI-risk controls (model risk, bias, auditability) from the start
- Track architecture debt and review decisions on their stated trigger, not the calendar

The pitfall: governance bolted on last. Retrofitted access, classification, risk, and audit are brittle and slow; designed in, they are enforced for free.

What good looks like. A readable decision log, standards teams can self-serve against, and risk controls that are part of the architecture rather than a gate at the end.

Illustrative frames: ADRs (Nygard) · UK Government Digital Service decision-record guidance · model-risk and AI-governance controls.

The maturity ladder

5 · Governed and self-funding capability

portfolio tied to capabilities; reference platform; decisions logged; risk designed in; AI pays for the next AI

4 · Architected and integrated

reference architecture; build/buy/boost decided on evidence; fitness functions; integrated to the systems of record

3 · Governed foundation

master data, contracts, and a semantic layer; ownership and lineage; pilots read trustworthy data

2 · Funded pilots, no architecture

initiatives running on borrowed data and point tools; no target architecture; each one re-solves the basics

1 · Scattered experiments

demos on laptops; no owner, no data foundation, no decision record

The four layers above are the climb from rung 2 to rung 5. Most organisations sit at 2 to 3 with a portfolio of pilots, and mistake activity for capability.

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Four failures that span the whole transformation

- **No target architecture.** A folder of pilots, each re-solving data, integration, and serving from scratch. Cost duplicates; nothing compounds.
- **The data foundation skipped.** Models built on ungoverned data inherit its errors; the score is fine in the pilot and wrong at scale.
- **Build-versus-buy by vendor pitch.** The biggest-budget decision made on a demo, with no weighted drivers, no cost model, and no account of the existing stack.
- **Governance retrofitted.** Security, risk, and audit added after the fact are brittle, slow, and the first thing a regulator or an incident exposes.

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The lens behind this view

My view comes from two vantage points usually held apart. One is over eighteen years directing enterprise data and analytics at scale across multiple ERP estates. There, master data, integration, the modelled layer, and the build-versus-buy call are real decisions with real budgets, not boxes on a diagram. The other is doctoral work on efficient, proactive AI, where honest evaluation and the cost of a wrong decision are the whole job.

The four layers above are where I have watched AI transformation get funded, and stall.

A worked example is public: a costed build/buy/boost architecture decision, with an ADR and a risk-adjusted model, at github.com/mtusman-ai/ai-build-buy-boost

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In one line

AI does not stall on the model; it stalls on the enterprise around it. Tie initiatives to capabilities, govern the data foundation, decide build-versus-buy on evidence, and design governance in. Scattered pilots become a capability that scales.

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Sources

WEF *Future of Jobs* (2025): AI and big-data roles fastest-growing to 2030 · Gartner (May 2024): ~48% of AI projects reach production, ~8 months prototype-to-production · Gartner (2024): ≥30% of generative-AI projects abandoned after proof of concept by end-2025 · Gartner (2026): forecast that 40% of enterprises demote or decommission their AI agents by end-2027 over governance gaps · RAND, *The Root Causes of Failure for AI Projects* (2024): 80%+ fail. Architecture practice draws on the enterprise-architecture literature: Architecture Decision Records (Nygard) and UK Government Digital Service decision-record guidance; *Data Management at Scale* (data contracts, semantic layer, data mesh); the MIT build-buy-partner framing; and architecture fitness functions (*Building Evolutionary Architectures*).

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See the full portfolio



The work behind this deck sits on one page: the Insights series, the projects and the code that runs them, and the publications.

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